

LEDGER · 2026-08-27 · BIOTECHNOLOGY / NANOMATERIALS

High-Porosity Vaterite Nanoparticles via Microbially Induced Calcium Carbonate Precipitation (MICP)

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Mesoporous Silica Nanoparticles (MSNs)
Entry application	Targeted Intravenous Drug Delivery Matrices
Measured advantage	none
Time to first revenue	24 months
Gross margin at parity	100.0%
Startup CapEx	\$355,000
Payback from first sale	0.0 months
Capital productivity	2738.03x
Regulatory risk	Moderate — qualification costs reported (\$750,000 estimated)

Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed, Negative control / sub-2x, Entry application mismatch, Form-factor / handling regression

Assessment

The concept explicitly concedes failure due to fatal aqueous thermodynamic instability (<48h calcite conversion causing burst release) and endotoxin contamination in IV delivery. Furthermore, startup CapEx (\$355,000) exceeds \$250,000 and velocity (24 months) exceeds the 18-month threshold.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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